EARTH'S FIELD NMR — a proton precession spectrometer

A coil round a bottle of water. POLARIZE: a few amperes through the coil for a few seconds magnetize the water's protons along the coil's field, a few hundred times more than the Earth's 46 uT alone would. Switch off: in a millisecond the field is gone, faster than the protons can follow, and they precess about the Earth's field at f = 42.58 Hz/uT × B: 1960 Hz here, a few microvolts in the same coil, dying away in a fraction of a second to a few seconds (T2*). LISTEN: the relay hands the coil to a tank tuned to f, and the preamplifier and the filters raise the microvolts to a volt.
The reference coil (handbook): 916 turns of AWG 20 on 3-inch pipe, 27 mH, 8.9 ohm, Q 37. At 1.5 A: 7.8 mT, about 1.2 uV in the coil from 1 L of water, 45 uV across the tank, 0.9 V at OUT on HIGH gain.

POLARIZING SUPPLY

6–30 V, up to 3 A. A bench supply in constant current sets the coil's current: or the coil's resistance does.

V POL: behind the 3 A fuse

J2 · F1 3A · D5 S534 · D6 SMBJ33A · C5 470uF · R3 15k · V POL

LED3 GREEN

LED3: the supply is there

COIL

A (2), B (1): the coil's two leads
S (3): its shield, if any
up to 150 V at switch-off:
do not touch
10–200 mH, up to 3 A, 1/2 L i^2 = 1 J

K1 HFD3/12: released, the coil LISTENS (as drawn);
energized, it is POLARIZED: A to V POL, B to the switch

J3 · COIL A · COIL B · K1B HFD3/12 · V POL

TANK+ · TANK−

D9 SMBJ12CA

D9: a clamp, idle unless a fault dumps the coil's energy here

DRAIN · SOURCE · Q1 IRF54227 · D7 SMCJ100A · R5 470 · R6 100k · R4 0.1R · I POL

I POL: 0.1 V per ampere, on and ramping off

D7 SMCJ100A holds the drain near 130 V while the coil's current falls to zero in a straight line (0.4 ms for the reference coil): faster than the protons can follow

GATE

CAL IN

a generator through 1M: I = V / 1M into the tank. mV-level, 5 Vpp max

J4 · R14 1M · R13 51R

SW1 TUNE

SW1 TUNE: each switch adds its capacitance (C0G) to the tank. In 3.3 nF steps up to 841 nF. Resonance: f = 1 / (2 pi sqrt(L C)): tune to the Larmor frequency for the most signal. The reference coil needs 246 nF at 1960 Hz. Caps C10–C29: 3.3n, 6.8n, 10n 0805, 47n, 100n 1206

gain 101 = 1 + 2 × 10k / 200R (between the two − inputs)

PREAMPLIFIER

an instrumentation amplifier, gain 101: U1 and U2 amplify each side of the tank, U3 takes the difference, so pickup common to both cancels. OPA197: 5.5 nV/rtHz and no current noise against the tank's 12 k: 1.7 dB noise figure with the reference coil. R11, R12 (1M) give the inputs their DC: R15, R16 (1k) limit a fault's current into them

R11 1M · R15 1k · U1 1 · R17 10k · INA A

R19 200R

R18 10k · U2 2 · R16 1k · R12 1M · INA B

R21 10k · R22 10k · R20 10k · R23 10k · U3 1 · PREAMP

HIGH–PASS

4th-order Butterworth, 688 Hz: two unity-gain Sallen-Key sections, Q 0.55 at 726 Hz and Q 1.31 at 670 Hz. 50 Hz −85 dB, 180 Hz −47 dB, 420 Hz −17 dB: mains hum and its low harmonics go. C0G capacitors.

C30 10nF · C31 10nF · R24 20k · R25 24k · U4 1 · HP1

LOW–PASS

2nd-order Butterworth, 4097 Hz (Q 0.72): the top of the band, and the anti-alias filter for a sound card. C0G capacitors.

C32 10nF · C33 10nF · R26 9.1k · R27 62k · U5 1 · HP

R28 8.2k · R29 8.2k · C34 6.8nF · C35 3.3nF · U6 1 · LP

GAIN

×21 or ×201 = 1 + 200k / Rg

U7 2 · R30 200k · R31 10k · R32 1k · SW2 GAIN
1: LOW, ×21.21 overall (67 dB)
3: HIGH, ×201.01 (86 dB)

OUT

the FID, about 1 V on HIGH:
1k in series: for a scope, an ADC or a sound card's line input

OUT 1k · R33 1k · J5

HEADPHONES

a mono copy of OUT for a stereo headset: 30 mV pf FID at a 32 ohm earpiece on HIGH. the knob full up (about 81 dB SPL): R56, R57 hold a mono plug's shorted ring to 21 mA. Jh Same Sky SJ1-3523N. 1 sleeve, 2 tip, 3 ring

R58 62k · RV3 10k · U7 1 · PHONES · R56 100R · R57 100R · J9

VOLUME clockwise louder

POWER

+15 V, GND, −15 V, 100 mA a rail

J1 · D1 S534 · D3 S534 · D4 S534 · D2 S534 · C1 100uF · C2 100uF · C3 100nF · C4 100nF · R1 10k · R2 10k · LED1 GREEN · LED2 GREEN · +15V · −15V · PWR_FLAG

D1, D2: a reversed supply does nothing
D3, D4: clamp a rail left open
LED1, LED2: both rails present

SEQUENCER

POL CMD high = POLARIZE; the relay pulls in at once, and the switch closes 18 ms later (U9).
POL CMD low: the switch opens at once, the coil's current ramps off through D7, and the relay lets go 15 ms later (U10): LISTEN.

RV1 POLARIZE TIME
0.2–4.7 s, clockwise longer

RV2 LISTEN TIME
0.2–4.8 s, clockwise longer

R40 100k · RV1 2M · RV2 2M · D10 1N4148W · R41 100k · D11 1N4148W · C50 1u · C51 1u · C52 1u · U8 1 NE555 · TIMER

C50–C52: 3 × 1 uF polypropylene film; t = 0.75 R C each way

SW3 SEQUENCE
1: EXT, the POLARIZE IN jack
3: INT, the timer (RV1, RV2)

SW4 MODE
1: RUN
3: LISTEN ONLY (no polarizing)

R42 100k · POL CMD

POLARIZE IN

high (> 1.3 V = polarize: TTL, 3.3 V, 5 V, a function generator: +/−30 V max; open = listen

J6 · R43 100k · D12 BAV99 · R44 10k · R45 100k · R46 10k · U8 2 · R47 1M · R48 10k · EXT

U8: a comparator at 1.3 V (R45, R46), 147 mV of hysteresis (R47)

R49 DELAY 220k · D13 1N4148W · C55 100nF · U9 1 · R7 4.7k · GATE

U9 GATE: on 18 ms after POL CMD rises (R49, C55), off at once when it falls (D13)

R50 HOLD 270k · D14 1N4148W · C56 100nF · U10 1 · R9 10k · RELAY

U10 RELAY: on at once (D14), off 15 ms after POL CMD falls (R50, C56)

K1 coil 1120 ohm · D8 1N4148W · R10 3.3k · LED4 RED · red: POLARIZING

R8 10k · Q2 MMBT3904

R51 10k · R52 10k · 7.3 V · C54 100nF

R53 10k · Q3 MMBT3904 · R55 1k · R54 10k · J8

SYNC OUT

follows POL CMD: 0 / 14 V through 1k: its falling edge is the switch-off: trigger the scope on it

GND · ground loops for the scope's clip

MH1 MH2 MH3 MH4

M3 mounting holes, one per corner

Bypass: 100 nF X7R at every op-amp's, comparator's and the timer's supply pins (C80–C99)

Open source, CERN-OHL-P-2.0 — github.com/gallicchio/earth-nmr-pcb
OPA197 for every amplifier, TS391 comparators, NE555 timer, IRFS4227 switch, HFD3 relay
Larmor frequency 1960 Hz at Harvey Mudd College, Claremont CA (46.0 uT, IGRF-14, 2026-11-01): tuned
Jason Gallicchio and Claude

Sheets: /
File: earthnmr.kicad_sch

Title: Earth's Field NMR: proton precession in the Earth's field

Size: A1	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/1